

Sweetened and water solutions consumption in stevia and sucrose groups.

To determine whether the pattern of fluid intake differed according to the age of exposure, body-weight-normalized fluid intake (mL/g BW/day) was analyzed using a four-way repeated-measures ANOVA, with age (juvenile or adult) and treatment (sucrose or stevia) as between-subject factors and fluid type (water or sweetened solution) and time as within-subject factors. The cage was considered the experimental unit. Because juvenile and adult animals were exposed at different postnatal ages, time points were aligned according to the corresponding day of treatment.

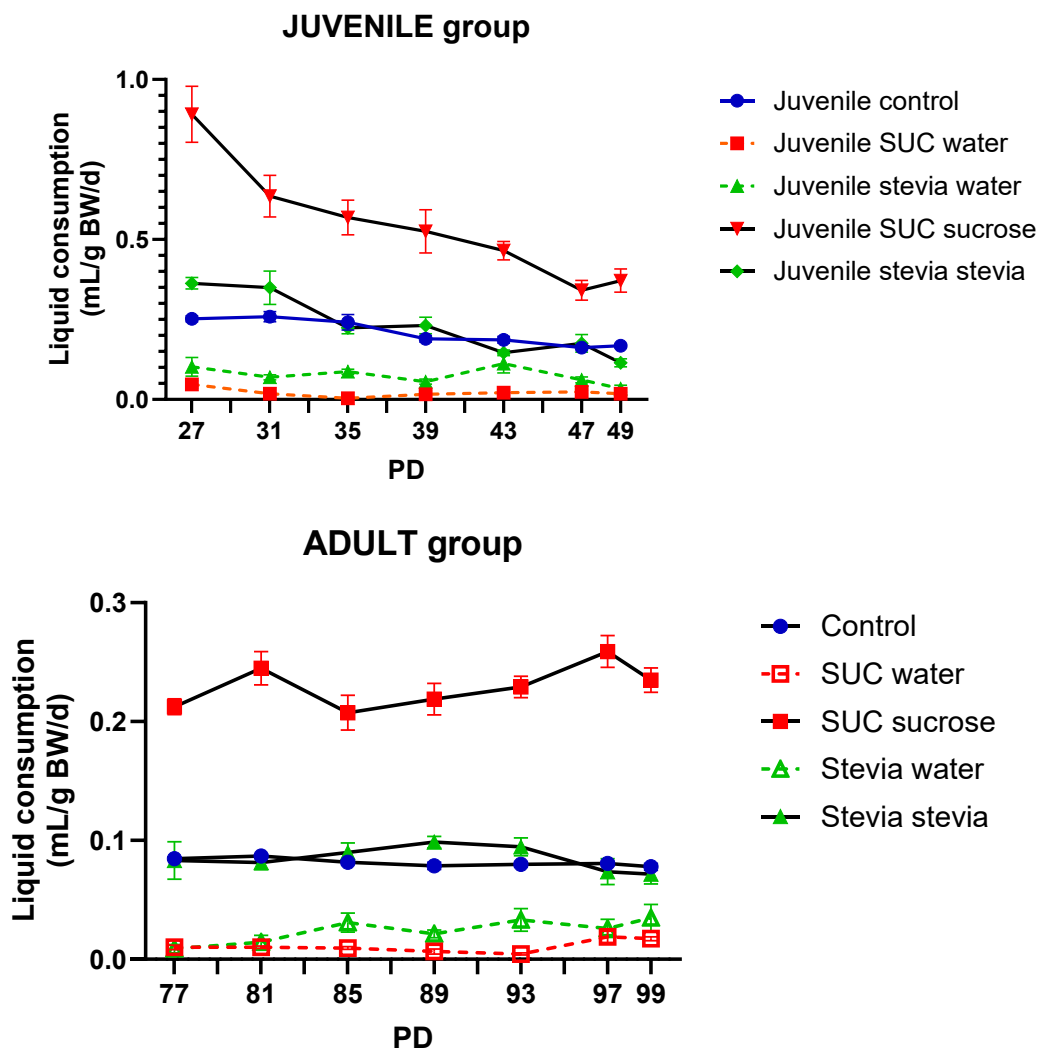
A significant age $\times$ treatment $\times$ fluid $\times$ time interaction was detected [$F(2.51, 60.21) = 7.97, p = 0.0003$], indicating that the temporal pattern of water versus sweetened-solution intake differed according to both the age of exposure and the type of sweetened beverage. A significant age $\times$ treatment $\times$ time interaction was also observed [$F(2.89, 69.27) = 6.98, p = 0.0004$].

When consumption was averaged across the experimental period, a significant age $\times$ treatment $\times$ fluid interaction was observed [$F(1, 24) = 11.99, p = 0.0020$]. Juvenile animals showed substantially greater body-weight-normalized intake of both sucrose and stevia solutions than adult animals (Fig. 1C). Stevia solution intake was approximately 2.8-fold higher in juveniles than in adults (0.241 ± 0.026 vs. 0.086 ± 0.002 mL/g BW/day), whereas sucrose intake was approximately 2.4-fold higher (0.538 ± 0.044 vs. 0.229 ± 0.008 mL/g BW/day). However, the relationship between sweetened-solution and water intake differed according to the type of sweetener. Sucrose intake remained consistently higher than water intake throughout the entire exposure period in both age groups. In contrast, stevia intake showed a more pronounced temporal modulation relative to water intake, particularly in juvenile animals, in which stevia intake was significantly higher than water intake from PD27 to PD41 but not from PD43 onward.

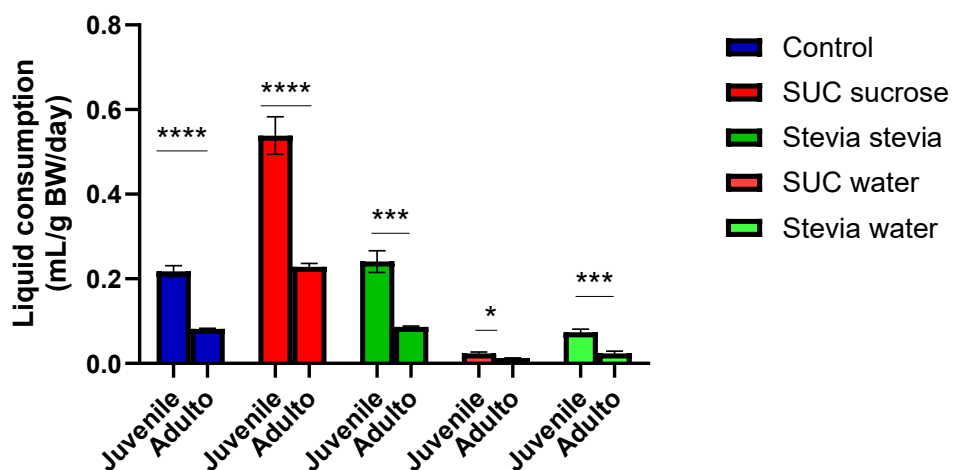
Together, these findings indicate that age influences not only the magnitude of body-weight-normalized sweetened-beverage intake but also its temporal pattern, with sucrose maintaining a more persistent predominance over water than stevia across age groups.

Figure 1. Fluid intake in juvenile and adult rats during the experimental period. (A) Time course of water and sweetened-solution intake in juvenile rats. (B) Time course of water and sweetened-solution intake in adult rats. Data in (A) and (B) were analyzed by repeated-measures ANOVA, followed by Šídák's multiple-comparisons test to compare sweetened-solution and water intake at each time point. (C) Mean fluid intake averaged across the experimental period in juvenile and

adult rats. Data are expressed as mean $\pm$ SEM ($n = 7$ cages/group). In (C), asterisks indicate age-related differences between juvenile and adult animals within each fluid condition. $p < 0.05$, $*p < 0.01$, $**p < 0.001$, and $***p < 0.0001$.



consumption averaged across the experimental period



Comparación Juvenile vs Adult p ajustado aprox. Asteriscos

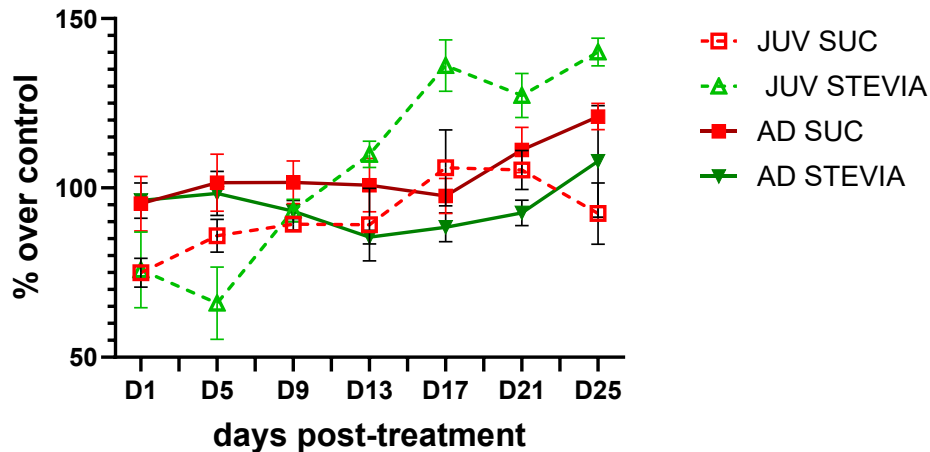
Control water	0.0000012	****
SUC sucrose	0.000084	****
Stevia stevia	0.00031	***
SUC water	0.031	*
Stevia water	0.00083	***

Consumption	Juvenile Mean	Juvenile SEM	n	Adult Mean	Adult SEM	n
Control – Water	0.2179	0.0131	7	0.0818	0.0012	7
SUC – Water	0.02364	0.00332	7	0.01158	0.00153	7
SUC – Sucrose	0.53839	0.04429	7	0.22862	0.00785	7
Stevia – Water	0.07347	0.00745	7	0.02391	0.00542	7
Stevia – Stevia	0.24081	0.02570	7	0.08609	0.00232	7

Water consumption after treatment

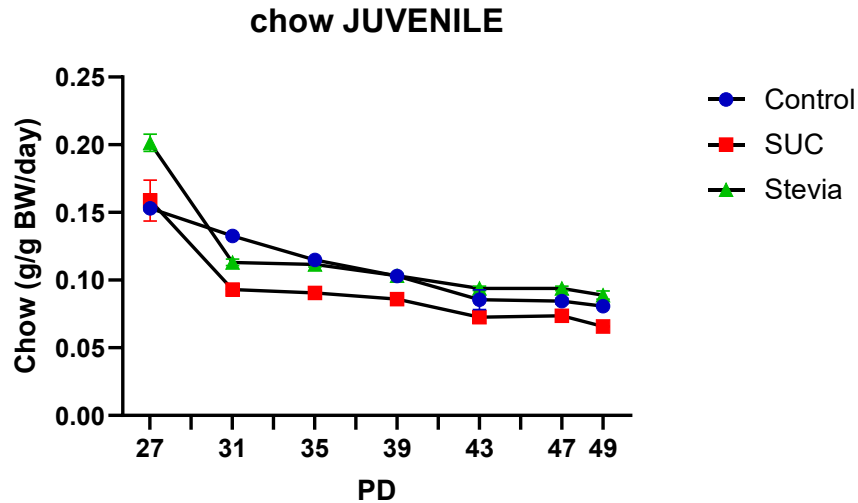
A three-way repeated-measures ANOVA, with age and treatment as between-subject factors and time as the within-subject factor, revealed a significant main effect of time [$F(4.59, 110.09) = 17.08$, $p < 0.0001$], as well as significant age $\times$ time [$F(4.59, 110.09) = 9.03$, $p < 0.0001$] and treatment $\times$ time interactions [$F(4.59, 110.09) = 3.29$, $p = 0.010$]. Importantly, a significant age $\times$ treatment $\times$ time interaction was also detected [$F(4.59, 110.09) = 7.58$, $p < 0.0001$], indicating that the temporal pattern of treatment effects differed according to the age of exposure. A significant age $\times$ treatment interaction was also observed [$F(1, 24) = 6.30$, $p = 0.019$], whereas no overall main effects of age [$F(1, 24) = 0.01$, $p = 0.919$] or treatment [$F(1, 24) = 0.14$, $p = 0.714$] were detected. Consistent with this interaction, consumption averaged across the experimental period showed an opposite pattern according to age: juvenile animals displayed mean values of 92.32 ± 5.58 and 105.67 ± 2.97 in the sucrose- and stevia-treated groups, respectively, whereas adult animals displayed mean values of 103.48 ± 5.44 and 93.57 ± 4.04 , respectively (mean $\pm$ SEM). The Geisser–Greenhouse correction was applied to effects involving the repeated-measures factor (time). Šídák's multiple-comparisons test showed that, in juvenile animals, stevia-treated rats exhibited higher values than sucrose-treated rats at D11 ($p = 0.030$) and D25 ($p = 0.005$), whereas no significant differences between treatments were detected at any time point in adult animals. When the effect of age was examined within the stevia-treated groups, juvenile animals showed

higher values than adults at D11 ($p = 0.014$), D13 ($p = 0.0016$), D15 ($p = 0.0003$), D17 ($p = 0.0018$), D21 ($p = 0.0075$), and D23 ($p = 0.031$). In contrast, no significant differences between juvenile and adult animals were detected at any time point within the sucrose-treated groups (all adjusted $p > 0.05$). These findings indicate a marked age-dependent divergence in the temporal response to stevia treatment, which became evident primarily during the second half of the exposure period.



Chow consumption

Chow intake was significantly affected by treatment in both juvenile [$F(2, 11) = 26.88$, $p = 0.0002$] and adult rats [$F(2, 12) = 33.99$, $p < 0.0001$]. A significant effect of time was also observed in both juvenile [$F(2.34, 21.07) = 203.97$, $p < 0.0001$] and adult animals [$F(1.51, 18.10) = 91.04$, $p < 0.0001$]. Moreover, a significant treatment $\times$ time interaction was detected in juveniles [$F(4.68, 21.07) = 11.54$, $p < 0.0001$] and adults [$F(3.02, 18.10) = 14.99$, $p < 0.0001$], indicating that the temporal pattern of chow intake differed among treatment groups at both ages. When chow intake was averaged across the experimental period, post hoc Šídák multiple-comparisons tests showed that sucrose-exposed juvenile rats consumed significantly less chow than both control ($p = 0.0006$) and stevia-exposed rats ($p = 0.0001$), whereas control and stevia-exposed juveniles did not differ ($p = 0.678$). A similar reduction in chow intake was observed in adult sucrose-exposed rats compared with both control ($p = 0.0008$) and stevia-exposed animals ($p < 0.0001$). In adults, however, stevia-exposed rats showed a modest but significant increase in chow intake compared with controls ($p = 0.027$).

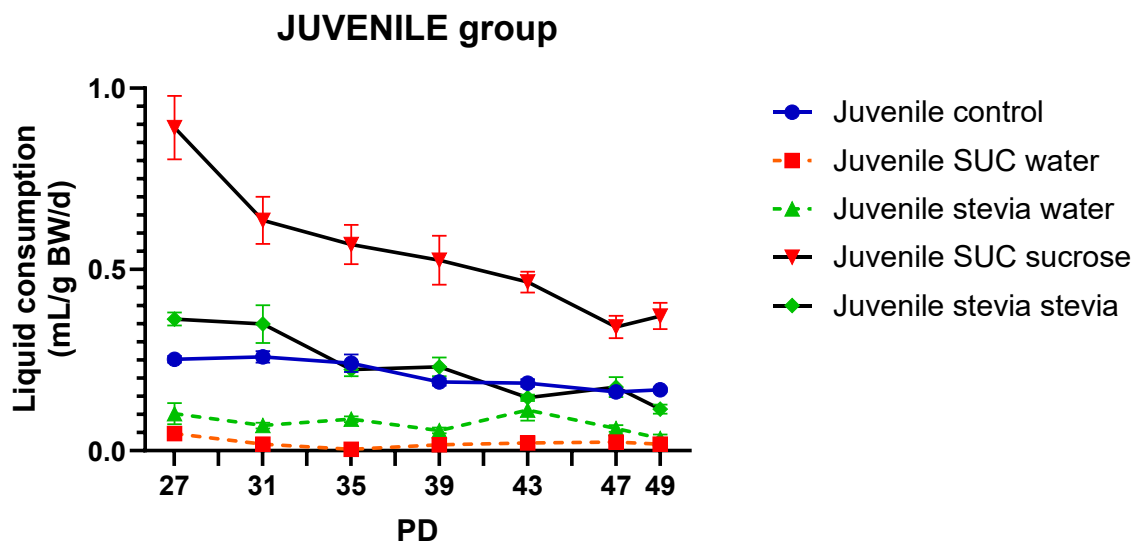


When chow intake was averaged across the experimental period and analyzed using a two-way ANOVA, significant effects of age [$F(1, 21) = 2363.85$, $p < 0.0001$] and treatment [$F(2, 21) = 57.37$, $p < 0.0001$], as well as a significant age $\times$ treatment interaction [$F(2, 21) = 3.86$, $p = 0.037$], were observed. Šídák's multiple-comparisons test showed that body-weight-normalized chow intake was significantly higher in juvenile than in adult animals in all treatment groups (Control, SUC, and Stevia; all $p < 0.0001$). Within juvenile animals, chow intake was significantly lower in the SUC group than in both Control ($p < 0.0001$) and Stevia ($p < 0.0001$), whereas Control and Stevia did not differ significantly. A similar reduction was observed in adult SUC-exposed animals compared with both Control ($p = 0.0005$) and Stevia ($p < 0.0001$). In adults, Stevia-treated animals showed a increase in chow intake compared with Control ($p = 0.0335$).

3-way RM ANOVA JUVENILE

Body-weight-normalized fluid intake (mL/g BW/day) was analyzed using a three-way repeated-measures ANOVA, with treatment (sucrose or stevia) as the between-subject factor and fluid type (water or sweetened solution) and time as within-subject factors. The analysis revealed significant main effects of treatment [$F(1, 12) = 25.44$, $p = 0.0003$], fluid type [$F(1, 12) = 157.00$, $p < 0.0001$], and time [$F(2.54, 30.52) = 32.68$, $p < 0.0001$].

In **JUVENILE animals** the three-way repeated-measures ANOVA, with treatment (sucrose or stevia) as the between-subject factor and fluid type (water or sweetened solution) and time as within-subject factors, revealed significant main effects of treatment [$F(1, 12) = 25.44$, $p = 0.0003$], fluid type [$F(1, 12) = 157.00$, $p < 0.0001$], and time [$F(2.54, 30.52) = 32.68$, $p < 0.0001$]. Significant treatment $\times$ fluid [$F(1, 12) = 40.73$, $p < 0.0001$], treatment $\times$ time [$F(2.54, 30.52) = 7.33$, $p = 0.0013$], and fluid $\times$ time [$F(2.22, 26.68) = 26.41$, $p < 0.0001$] interactions were also detected. Importantly, a significant treatment $\times$ fluid $\times$ time interaction was observed [$F(2.22, 26.68) = 7.32$, $p = 0.0022$], indicating that the relative consumption of water and sweetened solution changed over time in a treatment-dependent manner. Šídák's multiple-comparisons test showed that sucrose intake was significantly higher than water intake at all evaluated time points (PD27–PD49). In stevia-treated animals, stevia solution intake was significantly higher than water intake from PD27 to PD41, whereas no significant differences between stevia solution and water consumption were detected from PD43 onward. These findings indicate a persistent preference for the sucrose solution throughout the exposure period, whereas the relative preference for the stevia solution progressively diminished toward the end of treatment.



Te había recomendado “consumption averaged across the experimental period” en lugar de “averaged values”, porque suena más natural y preciso en un paper.

Para una figura, usaría:

Average fluid intake (mL/g BW/day)

Y en el texto:

“When fluid intake was averaged across the experimental period...”

O, si querés enfatizar que está normalizado por peso:

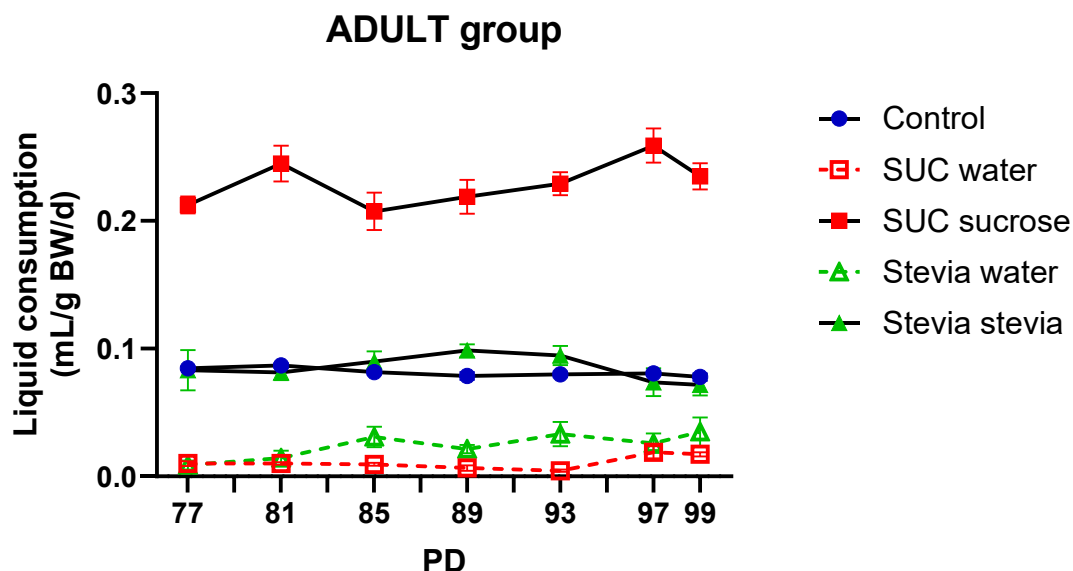
“Mean body-weight-normalized fluid intake across the experimental period...”

Para tu manuscrito, mi opción preferida es Average fluid intake (mL/g BW/day) para el eje Y/panel y consumption averaged across the experimental period en Results

3-way RM ANOVA ADULT

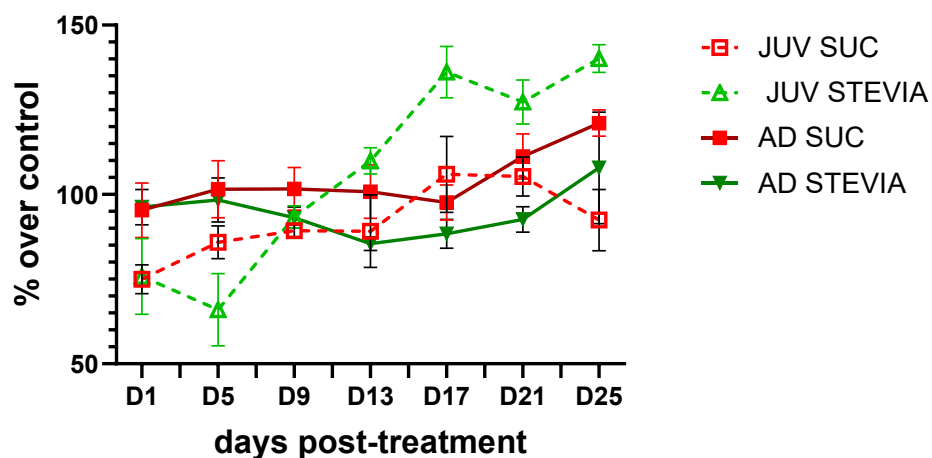
Within the **adult group**, body-weight-normalized fluid intake (mL/g BW/day) was analyzed using a three-way repeated-measures ANOVA, with treatment (sucrose or stevia) as the between-subject factor and fluid type (water or sweetened solution) and time as within-subject factors. Fluid consumption was normalized to the total body weight of the animals housed in each cage, with the cage considered the experimental unit. The analysis revealed significant main effects of treatment [$F(1, 12) = 246.88, p < 0.0001$] and fluid type [$F(1, 12) = 604.73, p < 0.0001$], as well as a significant treatment $\times$ fluid interaction [$F(1, 12) = 186.00, p < 0.0001$]. No significant main effect of time was detected [$F(4.03, 48.39) = 1.61, p = 0.187$]. A significant treatment $\times$ time interaction was observed [$F(4.03, 48.39) = 4.02, p = 0.0067$], whereas the fluid $\times$ time interaction was not significant [$F(2.85, 34.25) = 1.15, p = 0.342$]. Importantly, the treatment $\times$ fluid $\times$ time interaction was also not significant [$F(2.85, 34.25) = 2.00, p = 0.135$], indicating that, in adult animals, the relative consumption of water and sweetened solution did not show the marked treatment-dependent temporal divergence observed in juvenile animals.

Šídák's multiple-comparisons test showed that sucrose intake was significantly higher than water intake at all evaluated time points (PD77–PD99). In stevia-treated adult animals, stevia solution intake was significantly higher than water intake at PD77, PD79, PD81, PD83, PD87, PD89, PD91, and PD93, whereas no significant differences were detected at PD85 or from PD95 onward.

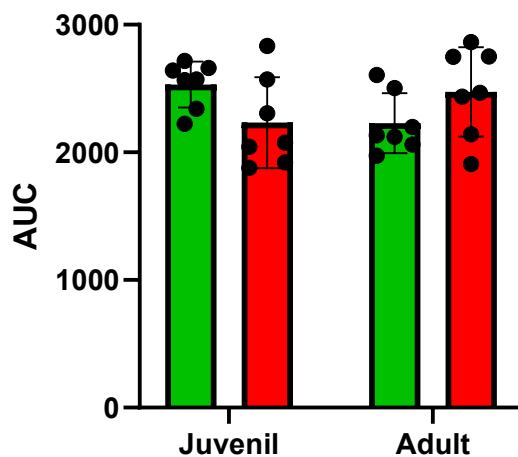


Water consumption after treatment (opcion 1):

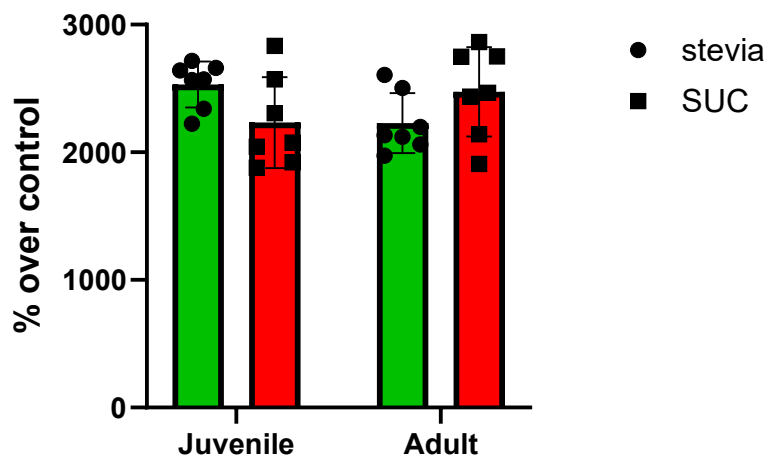
A three-way repeated-measures ANOVA, with age and treatment as between-subject factors and time as the within-subject factor, revealed a significant main effect of time [$F(4.59, 110.09) = 17.08, p < 0.0001$], as well as significant age $\times$ time [$F(4.59, 110.09) = 9.03, p < 0.0001$] and treatment $\times$ time interactions [$F(4.59, 110.09) = 3.29, p = 0.010$]. Importantly, a significant age $\times$ treatment $\times$ time interaction was also detected [$F(4.59, 110.09) = 7.58, p < 0.0001$], indicating that the temporal pattern of treatment effects differed according to the age of exposure. A significant age $\times$ treatment interaction was also observed [$F(1, 24) = 6.30, p = 0.019$], whereas no overall main effects of age [$F(1, 24) = 0.01, p = 0.919$] or treatment [$F(1, 24) = 0.14, p = 0.714$] were detected. The Geisser–Greenhouse correction was applied to effects involving the repeated-measures factor (time). Šídák's multiple-comparisons test showed that, in juvenile animals, stevia-treated rats exhibited higher values than sucrose-treated rats at D11 ($p = 0.030$) and D25 ($p = 0.005$), whereas no significant differences between treatments were detected at any time point in adult animals. When the effect of age was examined within the stevia-treated groups, juvenile animals showed higher values than adults at D11 ($p = 0.014$), D13 ($p = 0.0016$), D15 ($p = 0.0003$), D17 ($p = 0.0018$), D21 ($p = 0.0075$), and D23 ($p = 0.031$). These findings indicate a marked age-dependent divergence in the temporal response to stevia treatment, which became evident primarily during the second half of the exposure period. In contrast, no significant differences between juvenile and adult animals were detected at any time point within the sucrose-treated groups (all adjusted $p > 0.05$).



Analysis of the area under the curve (AUC) by two-way ANOVA revealed no significant main effects of age [$F(1,24) = 0.08$, $p = 0.781$] or treatment [$F(1,24) = 0.06$, $p = 0.810$]. However, a significant age $\times$ treatment interaction was detected [$F(1,24) = 6.13$, $p = 0.021$], indicating that the overall consumption pattern associated with each treatment differed according to the age of exposure.



Averaged RM values (better to summary the data)



Averaged values from the RM ANOVA:

Efecto	F(1,24)	p
Age	0.010	0.919
Treatment	0.138	0.714
Age × Treatment	6.299	0.0192

Otra manera de decirlo considerando los averaged values (ver que me gusta más).

Opcion 2:

A three-way repeated-measures ANOVA, with age and treatment as between-subject factors and time as the within-subject factor, revealed a significant main effect of time [$F(4.59, 110.09) = 17.08$, $p < 0.0001$], as well as significant age × time [$F(4.59, 110.09) = 9.03$, $p < 0.0001$] and treatment × time interactions [$F(4.59, 110.09) = 3.29$, $p = 0.010$]. Importantly, a significant age × treatment × time interaction was also detected [$F(4.59, 110.09) = 7.58$, $p < 0.0001$], indicating that the temporal pattern of treatment effects differed according to the age of exposure. A significant age × treatment interaction was also observed [$F(1, 24) = 6.30$, $p = 0.019$], whereas no overall main effects of age [$F(1, 24) = 0.01$, $p = 0.919$] or treatment [$F(1, 24) = 0.14$, $p = 0.714$] were detected. Consistent with this interaction, consumption averaged across the experimental period showed an opposite pattern according to age: juvenile animals displayed mean values of 92.32 ± 5.58 and 105.67 ± 2.97 in the sucrose- and stevia-treated groups, respectively, whereas adult animals displayed mean values of 103.48 ± 5.44 and 93.57 ± 4.04 , respectively (mean $\pm$ SEM). The Geisser–Greenhouse correction was applied to effects involving the repeated-measures factor (time). Šídák's multiple-comparisons test showed that, in juvenile animals, stevia-treated rats exhibited higher values than sucrose-treated rats at D11 ($p = 0.030$) and D25 ($p = 0.005$), whereas no significant differences between treatments were detected at any time point in adult animals. When the effect of age was examined within the stevia-treated groups, juvenile animals showed higher values than adults at D11 ($p = 0.014$), D13 ($p = 0.0016$), D15 ($p = 0.0003$), D17 ($p = 0.0018$), D21 ($p = 0.0075$), and D23 ($p = 0.031$). In contrast, no significant differences between juvenile and adult animals were detected at any time point within the sucrose-treated groups (all adjusted $p > 0.05$). These findings indicate a marked age-dependent divergence in the temporal response to stevia treatment, which became evident primarily during the second half of the exposure period.

Sucrose preference

Early-life exposure to stevia or sucrose did not significantly affect sucrose preference compared with the control group, either during the juvenile period or in adulthood. Two-way ANOVA revealed no significant main effect of treatment ($F(2,31) = 0.624, p = 0.543$) or age ($F(1,31) = 0.764, p = 0.389$), and no significant treatment $\times$ age interaction ($F(2,31) = 1.238, p = 0.304$). These findings indicate that early-life consumption of either stevia or sucrose did not induce persistent changes in sucrose preference and that preference remained comparable across the two age windows evaluated.